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PTX10002-36QDD Hardware Deepdive

An HPE Juniper Networking Techpost

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1 Introducing PTX10002-36QDD

Nicolas Fevrier - 03/19/2024

The new Juniper PTX10002-36QDD is here. It's our first 800GigabitEthernet, deep-buffer, high-scale, router in the market, powered by Express 5. And we are very excited to share some details about this unique platform.

1.1 Introduction/Overview

The PTX10002-36QDD is a compact form-factor router (2 Rack Units) offering 28.8Tbps of connectivity, and forwarding capability, with 36x 800GigE, 72x 400GigE or 288x 100GigE interfaces.

Based on the Express 5 Packet Forwarding Engine, it is designed to be deployed for multiple applications such as core, peering, CDN gateways, DCI, DC edge, aggregation, and datacenter, including AI/ML clusters.

This very high-performance router offers an impressive density of 800Gbps ports, as well as a range of advanced features that make it ideal for a variety of applications. Its deep buffers, dynamic resource allocation, and high-scale routing capabilities (supporting over 10 million routes in the FIB) provide exceptional flexibility and performance.

Additionally, it offers flexible filtering, inline traffic sampling, and native support for cutting-edge technologies such as SRv6 and BIER, making it a truly versatile and powerful solution for even the most demanding networking needs.

Central to the PTX10002-36QDD's design is its commitment to energy saving. The system-on-the-chip architecture cuts the power consumption per gigabit per second by half, compared to its predecessor. Additionally, it supports unique power optimization features, enabling the selective shutdown of certain forwarding capabilities during periods of non-use, thereby further enhancing energy efficiency.

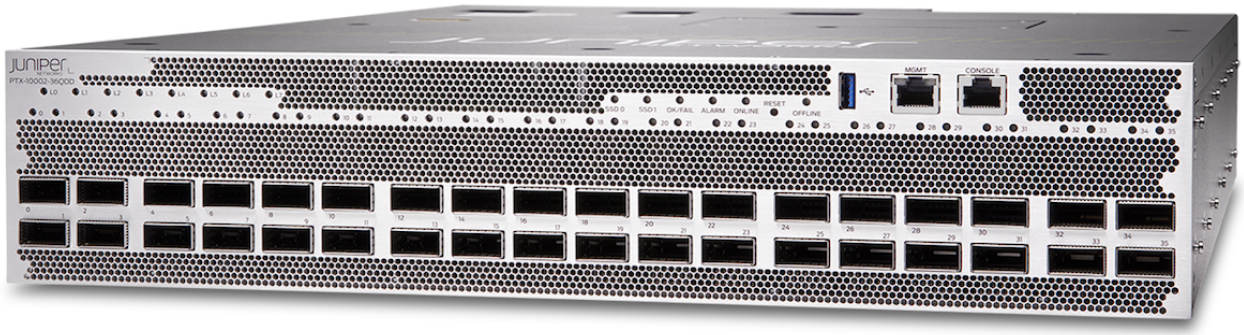


Figure 1: Side View of the PTX10002-36QDD

This platform is powered by the new Express 5, supporting MACsec at line rate on all ports, and class-C timing. Junos EVO is executed on a powerful multi-core CPU, enabling our multi-threaded routing stack, among other applications.

Let's now take a closer look at this router.

1.2 Main Forwarding Components

PTX10002-36QDD is the first router in our portfolio powered by the Express 5 Packet Forwarding Engine.

The Juniper Express silicon line, first introduced in 2011, aimed to transform the economics of packet transport networks by optimizing the forwarding path for density and interface speeds. Initially, the Express silicon line was designed for core routing functionality and relatively low scale, supporting MPLS Label Switch Router functions.

Over five generations, the Express chipsets evolved and expanded their capabilities, increasing the supported scale and extending their functionality.

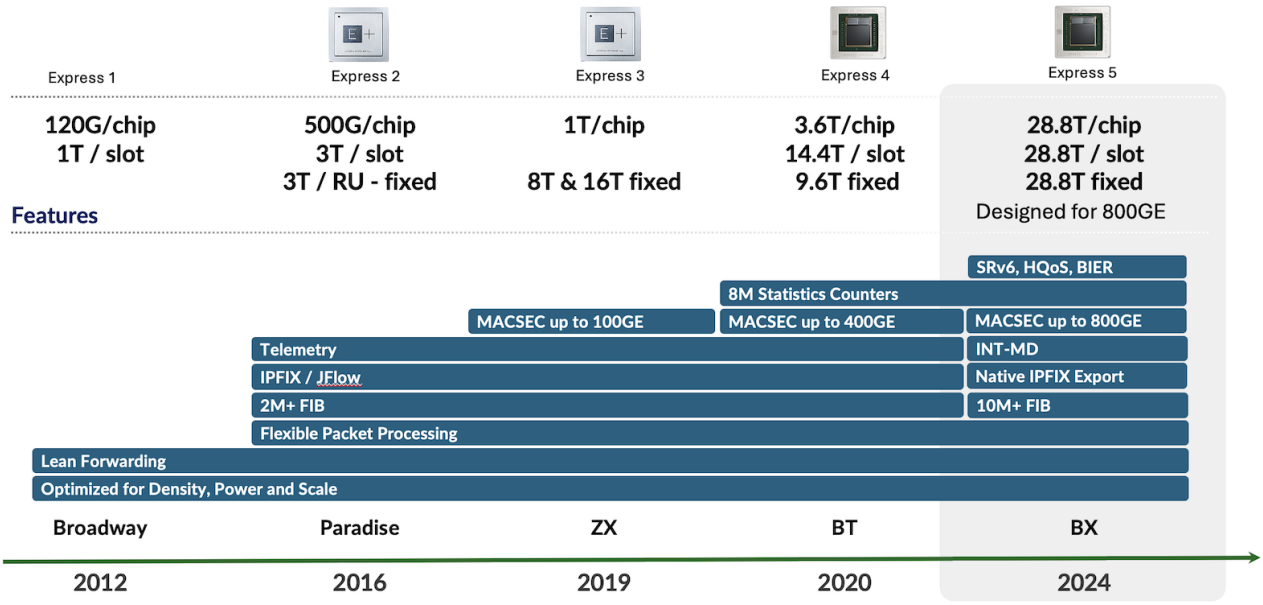


Figure 2: Express Packet Forwarding Engine Generations

Figure 2 above illustrates this evolution over generations. The Express 5 supports both high-scale and complex forwarding protocols and services like In-band Network Telemetry (INT-MD) and Hierarchical Quality of Service (H-QoS).

1.2.1 Express 5 High-Level Description

The Express 5 is not just a single chipset but more an ASIC family. It's the first in this industry to propose a design based on chiplets. You will find more details on this blog post from Dmitry Shokarev: https://techpost.hpe.com/articles/express-5-overview/article

To keep it short, we have two main chiplets, X and F, and we can mix and match them to create different packages based on the requirements.

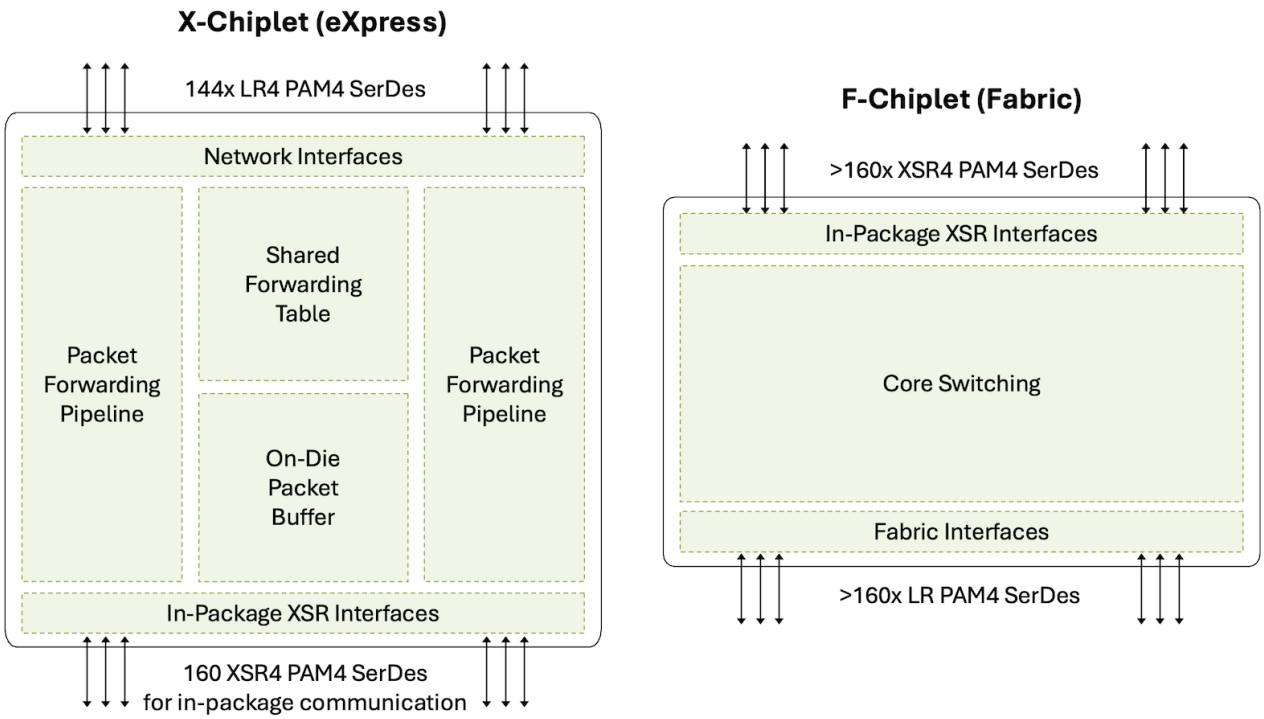


Figure 3: The Building Bricks of Express 5 ASIC Family: X-chiplet and F-chiplet

The X-chiplet offers 14.4Tbps of WAN SerDes on the north side (in Figure 3), supporting bandwidth from 10Gbps to 106Gbps. On the south part, they have very short reach links, XSR SerDes, to potentially interconnect to another X-chiplet or an F-chiplet, inside the same package.

The F-chiplet is dedicated to fabric connectivity. It offers interfaces on the south side of this diagram where cells will be received and transmitted to the fabric. And the north side is used to potentially interconnect to another X-chiplet or an F-chiplet, again in the same package.

These building bricks can be combined to meet our design needs. We can create packages made of:

- A single X-chiplet
- Two X-chiplets back-to-back, interconnected via the XSR4 SerDes. That's typically what we will use in the PTX10002-36QDD router. It's a 28.8Tbps ASIC and we name it BXX.
- One X-chiplet paired with one F-chiplet offers WAN connectivity on the network interface side and Fabric connectivity on the other side. We will find them in modular chassis line cards and they are called BXF.
- One F-chiplet, offering Fabric interfaces. You will find this configuration in modular chassis fabric cards, and they are called BF.

The PTX10002-36QDD is a standalone platform with no fabric connectivity requirements. It is based on a single BXX package made of two X-chiplets, presenting 288 WAN SerDes at 106Gbps. 288 electrical SerDes: that's 36 ports at 800Gbps or 72 ports at 400Gbps.

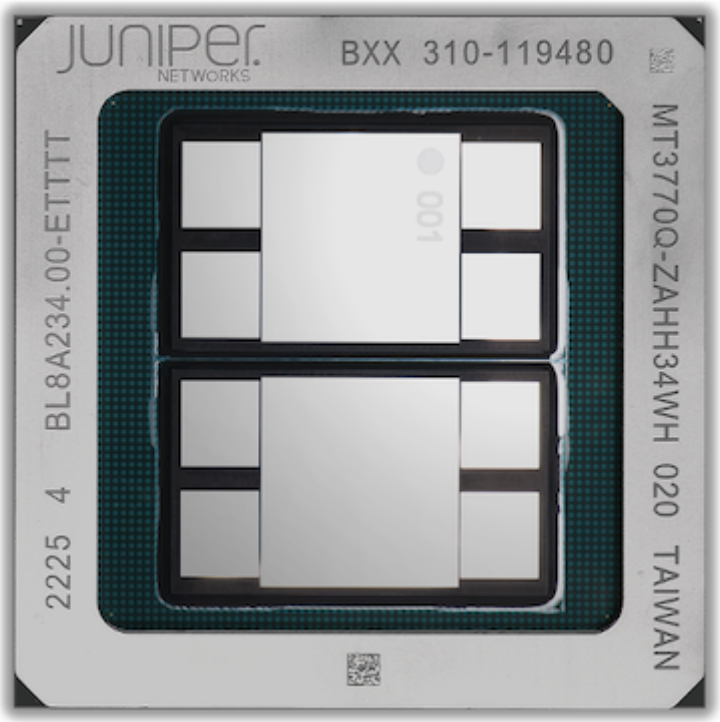


Figure 4: Express 5 Forwarding ASIC in BXX Configuration

With this package configuration, we don't waste resources for unused fabric. It's ideal for a SiP: System in the Package, where all interfaces are directly connected to the Forwarding ASIC. Since Express 5 supports MACsec internally, we won't need any additional parts to deliver MACsec encryption at line rate on all ports.

Table 1: Express 5 BXX Information

Metric	Value
Process Node	7nm
Internal Codename	BXX
WAN (Front Panel) Links	288x 106Gbps
Fabric (Internal) Links	None
Off-Chip Memory	32GB (4x 8GB) HBM
800GigE Port Density	36
400GigE Port Density	72
100GigE Port Density	288 (with 8x100GigE Breakout Cable)
50GigE Port Density*	288 (with 8x50GigE Breakout Cable)
40GigE Port Density	36
10GigE Port Density	144 (with 4x10GigE Breakout Cable)
Total Forwarding Capacity	28.8Tbps

Metric	Value
Total WAN Capacity	28.8Tbps
MACsec	Up to 800Gbps
Counters	8M
IPv4 or IPv6 FIB**	8M+

- Tested scale, hardware capable of much more (pending software validation)

FIB compression is enabled by default, but these numbers here represent FIB entries. In consequence, the scale could be much higher because of the compression. For more details on this technology, we invite you to check this article: https://techpost.hpe.com/articles/ptx-fib-compression/article

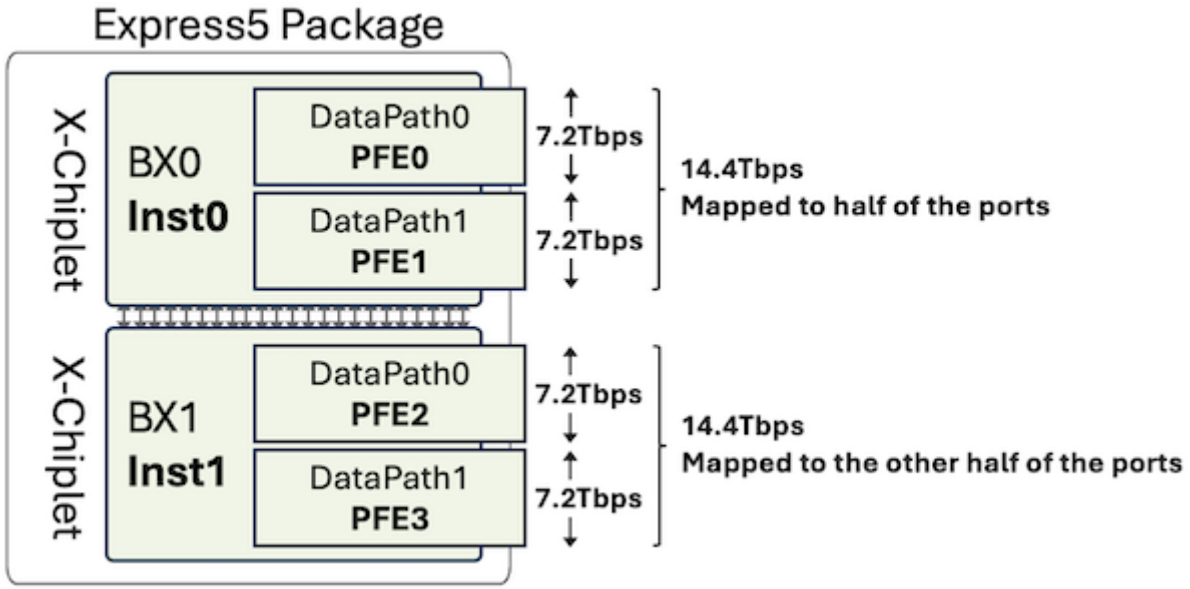


Figure 5: Express 5 BXX Chiplets and Datapaths

At a high level, we can represent the Express 5 Package used in PTX10002-36QDD as two PFE-instances (the chiplets) and four fully symmetric PFEs with identical throughput and packet performance (the datapaths or “slices” inside each chiplet die).

```
user@ptx10002-36qdd> show chassis fpc pfe all
FPC 0
PFE-Instance PFE PFE-State
0 0 ONLINE
0 1 ONLINE
1 2 ONLINE
1 3 ONLINE
user@ptx10002-36qdd>
```

We don’t need to remind you that any bandwidth figure expressed in this article is Full Duplex: a 7.2Tbps datapath can receive and transmit 7.2Tbps simultaneously.

For more details on the Express 5 design, capabilities, and differentiators, we invite you to check this article from Dmitry Shokarev: https://techpost.hpe.com/articles/express-5-overview/article

Now that we understand the forwarding engine used in this router, let's take a closer look at the PTX10002-36QDD.

1.2.2 Platform Architecture

The name of the PTX10002-36QDD is very much self-explanatory:

- PTX10k generation
- 2 Rack Units
- 36 Ports QSFP-DD 800

You can find the router datasheet here:

- https://www.juniper.net/us/en/products/routers/ptx-series/ptx10002-fixed-configuration-packet-transport-routers-datasheet.html
- https://www.juniper.net/us/en/products/routers/ptx-series/ptx10002-36qdd-packet-transport-router/specs.html

The usual "show chassis hardware details" provides a good overview of the internal parts.

```
regress@ptx10002-36qdd> show chassis hardware detail

Hardware inventory:
Item Version Part number Serial number Description
Chassis GBxxx PTX10002-36QDD [PTX10002-36QDD]
Midplane 0 REV 04 750-145729 BCDPxxxx Power Distribution Board
FPM 0 REV 03 750-141751 BCDKxxxx FPM-PTX10002
PSM 0 REV 09 740-073765 1GE2C15xxxx AC AFO 3000W PSU
PSM 1 REV 09 740-073765 1GE2C15xxxx AC AFO 3000W PSU
Routing Engine 0 BUILTIN BUILTIN RE-PTX10002-36QDD
  sda 200049 MB StorFly VSFDM4CC 58134-xxxx Solid State Disk
  sdb 200049 MB StorFly VSFDM4CC 58134-xxxx Solid State Disk
CB 0 REV 09 750-141519 BCDPxxxx Control Board
FPC 0 REV 03 750-141520 BCDRxxxx FPC-PTX10002-36QDD
  FPM 0 REV 03 750-153435 BCDVxxxx LED Board
  PIC 0 BUILTIN BUILTIN MRATE-36xQDD PIC
    Xcvr 0 REV 01 740-065630 1YCS042xxxx QSFP28-100G-AOC-1M
    Xcvr 6 REV 01 740-065630 1YCS042xxxx QSFP28-100G-AOC-1M
    Xcvr 8 REV 01 740-061405 1ECQ153xxxx QSFP-100GBASE-SR4
    Xcvr 12 REV 01 740-150870 1W1CSHA72xxxx QSFP-DD800-800G-AOC-1M
    Xcvr 35 REV 01 740-114884 UZLC4G0xxxx QSFP56-DD-400G-ZR
Fan Tray 0 REV 02 760-144017 BCDTxxxx PTX10002 Fan Tray, Front to Back Airflow - AFO
Fan Tray 1 REV 02 760-144017 BCDTxxxx PTX10002 Fan Tray, Front to Back Airflow - AFO
Fan Tray 2 REV 02 760-144017 BCDTxxxx PTX10002 Fan Tray, Front to Back Airflow - AFO

regress@ptx10002-36qdd>
```

1.2.3 Power and Cooling

The PTX10002-36QDD offers power supply 1+1 redundancy and front-to-back (AFO) fan trays with 2+1 redundancy.



Figure 6: Back View with Power Supply Modules and Fan Trays

The Power Supply Modules (PSM) exist in AC and DC but also in two flavors:

- 3,000W
- 2,200W

Depending on the PSM inserted, the system will operate in a different mode. We will cover this in detail in the Power Saving section, later in this article.

1.2.4 Block Diagram

You'll find below a schematic representation of the PTX10002-36QDD internal architecture:

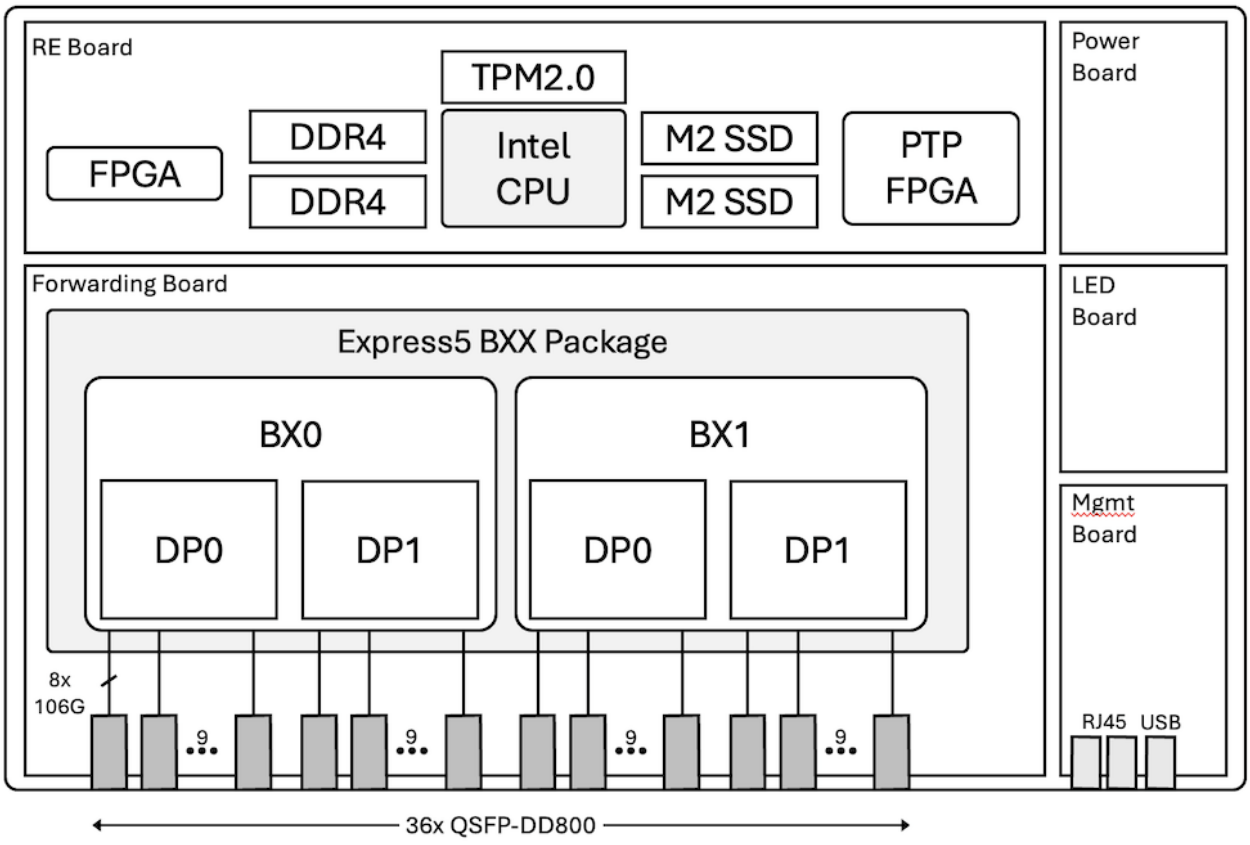


Figure 7: PTX10002-36QDD Block Diagram

This simplified hardware design contributes to the very competitive power consumption of the PTX10002-36QDD.

1.2.5 Interfaces

As illustrated in Figure 7, the architecture is very simple: we don't rely on any intermediate components such as Reverse GearBox (RGB) to connect the optical cages to the Packet Forwarding Engine Port Groups (PG).

The 36 ports of the router are split in half, with 18 ports mapped to BX0 and 18 other ports to BX1. In each chiplet, we have two datapaths (or slices) of 9 ports each.

From a router CLI perspective, these slices are named PFE0 to PFE3.

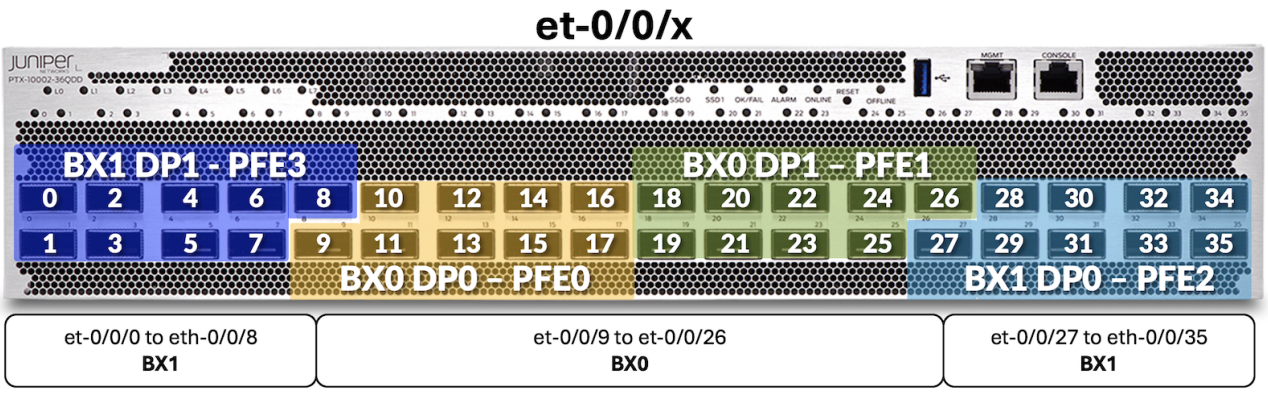


Figure 8: Port Mapping to Chiplet Dies and Datapaths

Consequently, the port naming format is also very simple:

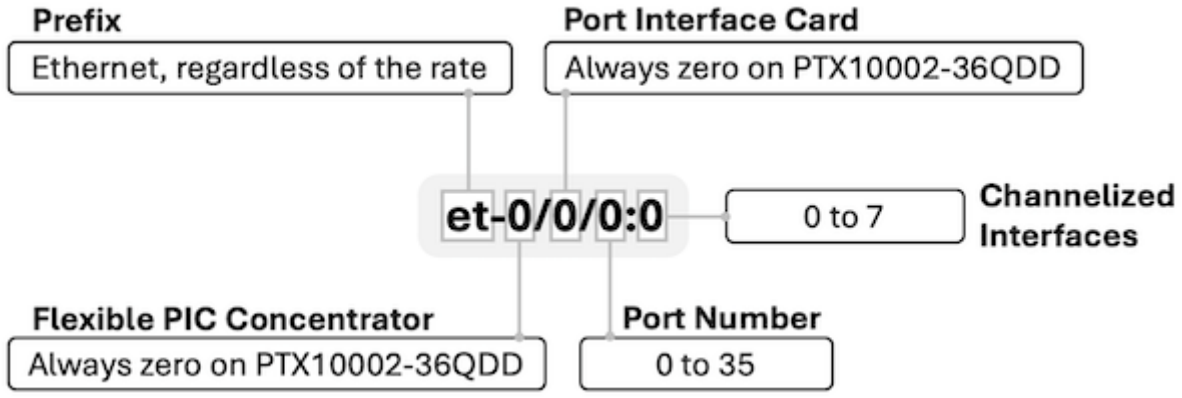


Figure 9: Port Naming Convention

Since the system is composed of a single FPC with a single PIC, all interfaces are named et-0/0/[0-35]:[0-7]. Only port number, and potentially channelization, are variable.

Interesting to note that 800Gbps ports will support new types of pluggable and connectors:

- QDD-2x400G, QDD-8x100G with Dual LC-Duplex
- And 8X100G interfaces with Dual MPO-12.

In both cases, there are 2 independent connectors (a pair) housed on a single transceiver which will simplify cable management:

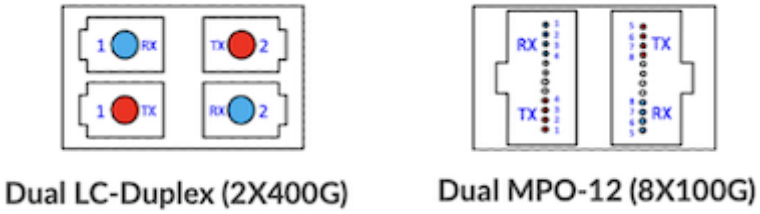


Figure 10: New 800G “Paired Connectors”

These connectors enable the high 400GigE port density without requiring specific breakout cables or patch panels: natively, we support 72x 400GigE ports on PTX10002-36QDD.

Each physical port is mapped to a unique port group (PG) via 8x SerDes and no intermediate RGB, so we don’t have any port combination limitation. You can use all ports with 2x400GigE or 8x100GigE without any constraints. Every port can be configured at the speed you need, but you can only support a unique speed for members of the same physical port channelized (4x10GigE, 4x25GigE, 8x100GigE, 2x400GigE).

The full list of supported interfaces will be updated soon on the pathfinder, the following chart provides a couple of examples.

Table 2: Interface rate and SerDes

Rate	Port Type	#SerDes and Rate (Gbps)
800GigE	1x 800GAUI-8	8x 106.25
400GigE	2x 400GAUI-4	4x 106.25
400GigE	1x 400GAUI-8	8x 53.125
200GigE	2x 200GAUI-4	4x 53.125
100GigE	8x 100GAUI-1	1x 106.25
100GigE	2x 100GAUI-4	4x 26.56254x 25.78125
50GigE	2x LAUI-2	2x 25.78125
40GigE	1x XLAUI	4x 10.3125
25GigE	4x 25GAUI-1	1x 25.78125
10GigE	1x XFI	1x 10.3125

As a quick on-box reference, the following CLI command shows transceivers plugged into the port, plus the port speed capabilities (note: it shows capabilities and not necessarily the software support. Please use the port checker and hardware compatibility tools on apps.juniper.net to verify the support).

```
regress@ptx10002-36qdd> show chassis pic fpc-slot 0 pic-slot 0
FPC slot 0, PIC slot 0 information:
  Type MRATE-36xQDD PIC
  State Online
```

```
  PIC version 255.09
  Uptime 15 days, 3 hours, 6 minutes, 26 seconds

PIC port information:
                    Fiber Xcvr vendor Wave- Xcvr JNPR MSA
  Port Cable type type Xcvr vendor part number length Firmware Rev Version
  0 100G AOC 1M MM JUNIPER-DELTA QAOC-100G4F1xxxx 850 nm 0.0 REV 01 SFF-8636 ver 2.7
  2 100G AOC 1M MM JUNIPER-DELTA QAOC-100G4F1xxxx 850 nm 0.0 REV 01 SFF-8636 ver 2.7
  6 100G AOC 1M MM JUNIPER-DELTA QAOC-100G4F1xxxx 850 nm 0.0 REV 01 SFF-8636 ver 2.7
  8 100GBASE SR4 MM JUNIPER-AVAGO AFBR-89CDDZ-xxx 850 nm 0.0 REV 01 SFF-8636 ver 2.7
  12 800G-AOC 1M MM JUNIPER-1W1 740-15xxxx 1311 nm 3.3 REV 01 CMIS 5.1
  35 400G-ZR SM JUNIPER-2E1 740-11xxxx 1528.77 nm - 1567.13 nm 1.6 REV 01 CMIS 5.0

Port speed information:

  Port PFE Capable Port Speeds
  0 NA 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  1 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
      2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  2 NA 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  3 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
      2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  4 NA 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  5 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
      2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  6 NA 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  7 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
      2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  8 NA 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  9 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
      2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  10 NA 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1
      x800G 2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  11 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  12 NA 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1
      x800G 2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  13 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  14 NA 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1
      x800G 2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  15 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  16 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  17 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  18 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  19 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  20 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  21 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  22 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  23 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  24 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  25 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  26 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  27 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  28 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  29 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  30 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  31 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
       2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
  32 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
```

```
          2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
54    33 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
          2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
55    34 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1x800G
          2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
56    35 NA 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 4x50G 8x25G 8x50G 2x100G 3x100G 4x100G 1x400G 1
         x800G 2x400G 5x100G 6x100G 7x100G 8x100G 4x200G
57
58  regress@ptx10002-36qdd>
```

1.3 Control Plane

PTX10002-36QDD is designed for demanding routing applications and is equipped with a powerful 10-core Intel CPU to support faster convergence and complex BGP policy processing. The table below lists the hardware specifications of the control plane parts.

Table 3: Control Plane Components

Component	Value
CPU	Intel Icelake-D, 10-core @ 3.0GHz
DRAM	128GB DDR4
Storage	SATA SSD 2x 200GB

The minimum support Junos release on PTX10002-36QDD is 23.4R2-S1-EVO.

Various Juniper software components leverage multi-core CPU capabilities. Impressive scale and performance are demonstrated: Routing Process Daemon supports multithreading capabilities to process routing updates and routing resolution. For example, the system learns BGP routes at the rate of 250k routes per second and updates the FIB at 42k routes per second.

Storage subsystem is comprised of solid-state drives (SSDs), two are provided for redundancy, plus reliable management of the software upgrades and rollbacks. These drives are not field-replaceable, but they can be removed from the system before shipping devices back to Juniper for service. This is to support customer's security policies where non-volatile storage may not leave customer premises. Removal of drives requires a non-standard service agreement. As an alternative to the physical removal, both SSDs support Secure Erase functionality.

PTX10002-36QDD control plane is designed to host 3rd party applications: there is enough storage, DRAM capacity, and CPU power. Some of these applications may include custom Service Assurance Agents or statistics collection agents developed by Juniper partners and customers. Even full Telegraf, InfluxDB, Grafana stack can run on the router itself for data collection and visualization, check out the blog post (https://techpost.hpe.com/articles/telemetry-collector-and-dataviz-on-junos-evo/article).

In terms of management plane connectivity, the CPU is directly linked to:

- the management port at 1Gbps
- each BX chiplet at 10Gbps

1.3.1 Timing

PTX10002-36QDD is designed to support timing: sync-E, and PTP class-C on all ports and even class-D quality for certain port configurations.

The router offers the following connectivity on the back of the chassis.

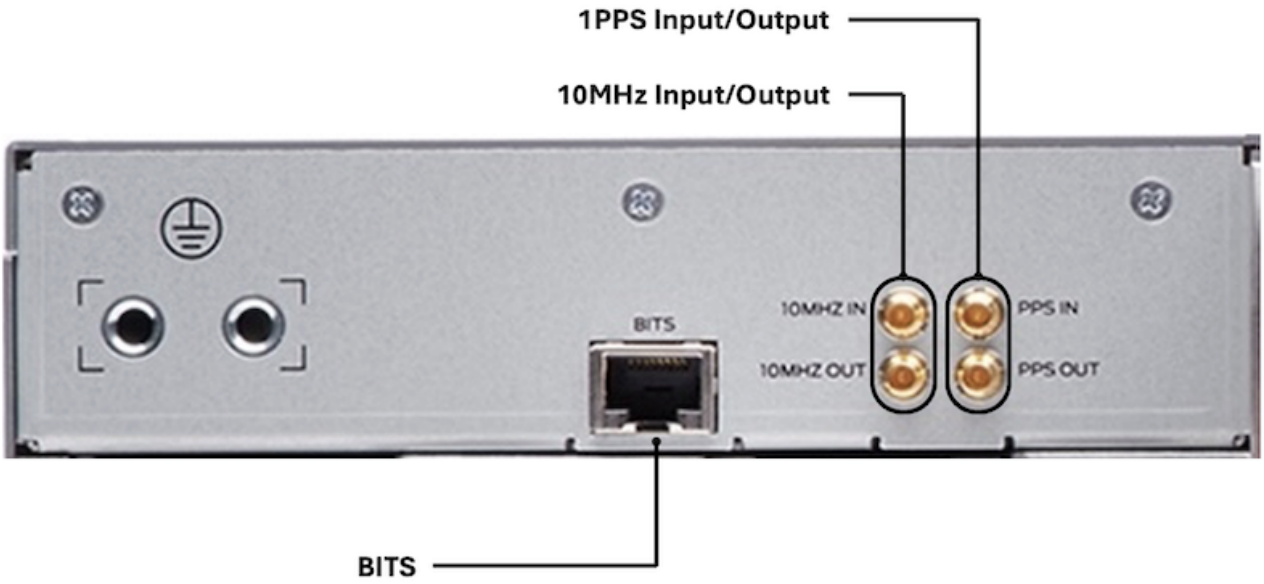


Figure 11: Timing Ports on the Rear Side of the Router

Internally, a dedicated timing circuitry is used, leveraging Juniper-developed PTP FPGA. Preliminary tests are showing even higher clock quality in certain cases. Stay tuned for more details.

1.3.2 Security

Image Validation

PTX10002-36QDD is equipped with a Trusted Platform Module (TPM2.0) located on the RE board. It makes it fully compliant with the TPM standard as published by the Trusted Computing Group (TCG).

The TPM's non-volatile storage is used as a persistent, access-controlled area for component registration, location of policy, keys, etc.

Trusted extensions are integrated into the Icelake CPU and only it can access the TPM.

Hardware and firmware are designed to support FIPS 140-2 Level 2.

MACsec

The MACsec services are handled in the MAC part of the Port Groups. That means we don't need to rely on any additional components (a retimer or reverse gearbox) to provide L2 encryption. It can be activated or deactivated at the port level and supports all operating speeds from 10Gbps to 800Gbps. When you don't configure the service, the MACsec part is clock-gated automatically, reducing power usage.

Firewall Filters

Express 5 filters are powerful and flexible with five lookup engines operating in parallel and a unique lookup key width of 736-bit.

1.4 Power Operation Modes and Energy Saving

We briefly introduced the different Power Supply Module options available in the PTX10002-36QDD:

- AC/HVAC/HVDC at 2200W
- DC at 2200W
- AC/HVAC/HVDC at 3000W
- DC at 3000W

Mixing 3000W and 2200W PSM in the same system is not supported.

The default behavior of the router will be different if you insert 3000W PSM or if you insert 2200W PSM. Let's explore these aspects now.

1.4.1 2200W PSM are Inserted

When these 2200W PSMs are present in PTX10002-36QDD, the system will automatically detect them and configure a "400G" mode also named "power-optimized" mode or sometimes "14.4Tbps" mode.



Figure 12: PTX10002-36QDD in 14.4Tbps / 400G Mode with 2200W PSM

In this operational mode:

- Configuration is forced by the PSM 2200W presence.
- Express 5 clock is reduced to 65% of its capability.
- Ports are "limited" to 400Gbps and global forwarding is 14.4Tbps.
- All configuration modes, such as 4x 112Gbps or 8x 56Gbps are supported.
- Insertion of an 800Gbps or 2x400Gbps optic is not supported.
- Total power consumption will be reduced by roughly 30%.
- Support up to 25W optics in each port (example 400GigE ZR).

1.4.2 3000W PSM are Inserted

Here, we have 3000W PSMs inserted in the back of PTX10002-36QDD, the system will automatically detect them and configure an "800G" mode also named "28.8Tbps" mode.



Figure 13: PTX10002-36QDD in 28.8Tbps / 800G Mode with 3000W PSM

In this operational mode:

- Configuration is forced by the PSM 3000W presence.

Note that it represents the PSM capacity, depending on the configuration the system will draw much less power. - Express 5 runs at full speed. - Ports can be used with at 400GigE, 2x400GigE, 800GigE, etc… - Support up to 30W optics in each port.

```
regress@ptx10002-36qdd> show chassis mode
         FRU Slot Mode
         FPC 0 Normal Mode

regress@ptx10002-36qdd> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 543
  PSM 0
    State: Online
    Input 1 203 253
    Output 12 195.09
    Capacity 3000 W (maximum 3000 W)
  PSM 1
    State: Online
    Input 1 198 290
    Output 12 243.03
    Capacity 3000 W (maximum 3000 W)

Item Used(W)
  Routing Engine 0 71
  FPC 0 332
  Fan Tray 0 10
  Fan Tray 1 9
  Fan Tray 2 9

System:
  Power mode: Normal
  Zone 0:
     Capacity: 6000 W (maximum 6000 W)
     Allocated power: 2625 W (3375 W remaining)
     Actual usage: 543 W
  Total system capacity: 6000 W (maximum 6000 W)
  Total remaining power: 3375 W

regress@ptx10002-36qdd>
```

1.4.3 3000W PSMs are Inserted and Power-Optimized Config is Committed

In this third option, the system detects 3000W PSMs, but we configured and committed “set chassis mode power-optimized”. After a reload. This configuration makes the system behave very similarly to the 14.4Tbps mode with 2200W PSMs.

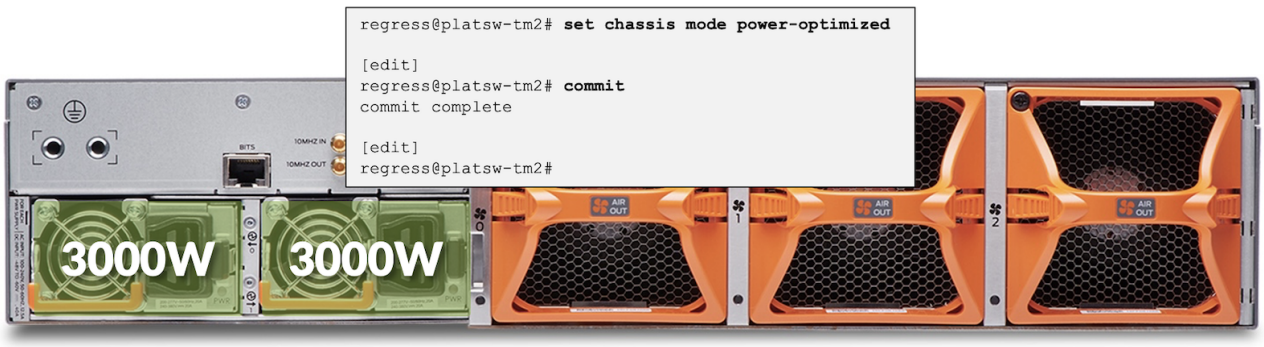


Figure 14: PTX10002-36QDD in Power-Optimized Mode with 3000W PSM

In this operational mode:

- Configuration is forced by CLI and reload.
- Express 5 speed is reduced to 65% of its capability.
- Ports are “limited” to 400Gbps and global forwarding is 14.4Tbps.
- Total power consumption will be reduced by roughly 30%.
- Support up to 25W optics in each port.

```
regress@ptx10002-36qdd> show chassis mode
         FRU Slot Mode
         FPC 0 Power Optimized Mode

regress@ptx10002-36qdd> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 317
  PSM 0
    State: Online
    Input 1 202 137
    Output 12 103.51
    Capacity 3000 W (maximum 3000 W)
  PSM 1
    State: Online
    Input 1 198 180
    Output 11.99 145.42
    Capacity 3000 W (maximum 3000 W)

Item Used(W)
  Routing Engine 0 85
  FPC 0 0
  Fan Tray 0 7
  Fan Tray 1 6
  Fan Tray 2 7

System:
  Power mode: power-optimized
  Zone 0:
     Capacity: 6000 W (maximum 6000 W)
     Allocated power: 2150 W (3850 W remaining)
     Actual usage: 317 W
  Total system capacity: 6000 W (maximum 6000 W)
  Total remaining power: 3850 W

regress@ptx10002-36qdd>
```

We performed the test in the lab and measured the following:

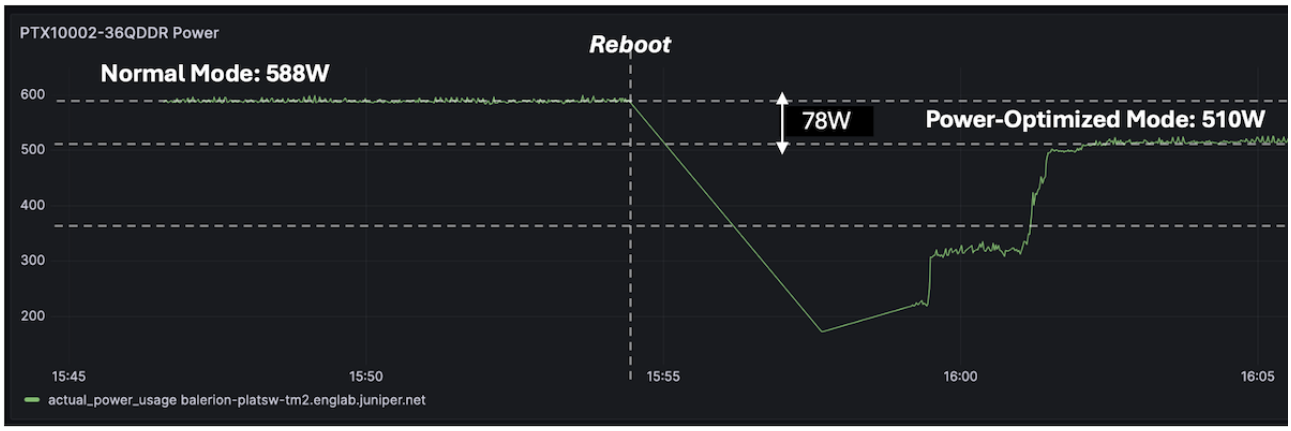


Figure 15: Configuring power-optimized mode and reload

In this quick test, we notice around 80W of power saving between the two modes after reboot. Please note your mileage may vary depending on the optics inserted and environmental conditions.

1.4.4 X-Chiplet Die Power Off

Since the Packet Forwarding Engine Express 5 powering the PTX10002-36QDD is a BXX design, it's possible to power off one of the chiplets. It will reduce the connectivity and the forwarding capability by half (to 18 ports and 14.4Tbps respectively), and will significantly reduce power usage.

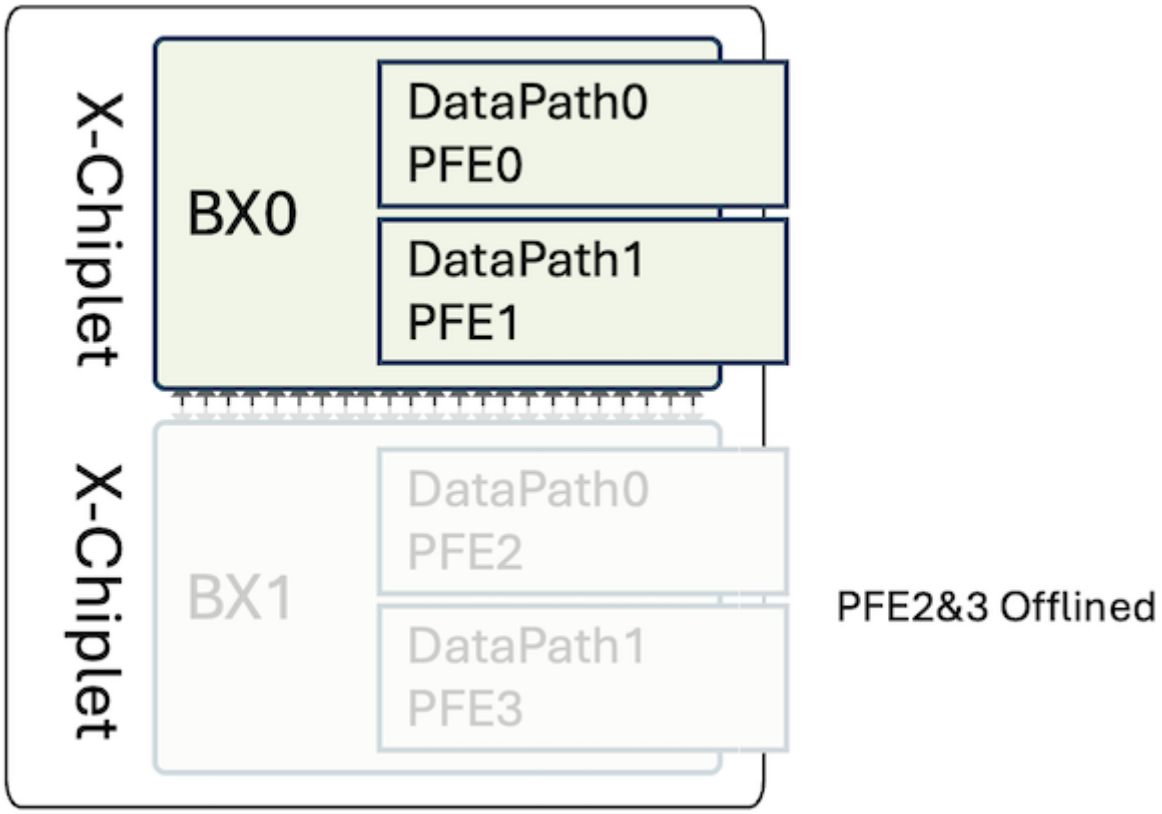


Figure 16: Express 5 BXX with one X-Chiplet Powered Off

It's possible to power on and off the die, at will. Note that, at FRS, the power on/off of the PFEs will stall the

routes and next-hop programming for approximately 30s.

The ports are directly mapped to the different datapaths/slices. Therefore, if we power off BX1, we will see 18 ports disappear from the IFD list (physical interfaces).

Table 4: Chiplet Die to Port Mapping

X-Chiplet	PFE-Instance	PFE Port Mappinget-0/0/x
BX0	PFE0 / PFE1	9/10/11/12/13/14/15/16/1718/19/20/21/22/23/2
BX1	PFE2 / PFE3	0/1/2/3/4/5/6/7/827/28/29/30/31/32/33/34/35

The “power off” configuration can be performed at two levels: in config mode, or through request chassis command. The element we will configure here is the 7.2Tbps datapath/slice, which is called “PFE” from the router’s perspective.

In CLI mode, we just need to request to “offline” one PFE of the pair (number 2 in our example), and it will automatically apply it to the second PFE (number 3 here). In the output below, we tried to offline the PFE3 and the system refused since this slice was already in the process of being shutdown.

```
regress@ptx10002-36qdd> show chassis fpc pfe all
FPC 0
PFE-Instance PFE PFE-State
0 0 ONLINE
0 1 ONLINE
1 2 ONLINE
1 3 ONLINE
regress@ptx10002-36qdd> request chassis fpc slot 0 pfe 2 offline
Warning: PFE 3 will also be offlined by this command. Proceed? [yes,no] (no) yes
Fru Block offline initiated!
regress@ptx10002-36qdd> request chassis fpc slot 0 pfe 3 offline
Warning: PFE 2 will also be offlined by this command. Proceed? [yes,no] (no) yes
Fru Block cannot be offlined. Already in state TRANSITION_OFFLINE
regress@ptx10002-36qdd> show chassis fpc pfe all
FPC 0
PFE-Instance PFE PFE-State
0 0 ONLINE
0 1 ONLINE
1 2 Offlined by CLI
1 3 Offlined by CLI
regress@ptx10002-36qdd>
```

In configuration mode, you will need to specifically power off both PFEs of the pair, otherwise Junos will refuse the commit.

```
regress@ptx10002-36qdd# set chassis fpc 0 pfe ?
Possible completions:
  <pfe-id> PFE(Packet forwarding engine) identifier (0..3)
[edit]
regress@ptx10002-36qdd# set chassis fpc 0 pfe 2 power off
[edit]
regress@ptx10002-36qdd# commit
[edit chassis fpc 0 pfe 2 power]
  'power off'
    pfe 2,3 need to be configured power on / off together. Config for these pfes need to be added
         / deleted / changed together.
error: commit failed: (validation hook evaluation failed)
[edit]
regress@ptx10002-36qdd# set chassis fpc 0 pfe 3 power off
[edit]
regress@ptx10002-36qdd# commit
commit complete
[edit]
regress@ptx10002-36qdd# run show chassis fpc pfe all
FPC 0
```

```
PFE-Instance PFE PFE-State
0 0 ONLINE
0 1 ONLINE
1 2 Configured power off
1 3 Configured power off
[edit]
regress@ptx10002-36qdd#
```

In our lab, we ran this quick test and measured the power saved in the process:



Figure 17: Turning off and on one PFE-instance

In this test above, we turn off BX1 and turn it back on. We notice around 150W of power saved here. Note that we have optics inserted in the ports mapped to BX0 but no optics were present on the BX1 side. Again, it's a lab exercise performed in a non-controlled environment and your mileage may vary a bit in other conditions. Nevertheless, it gives a good approximation of the savings we can expect.

You can also mix the two different approaches described above, meaning: you can enable the power-optimized mode (reducing the PFE clock and limiting the ports to 400Gbps) **and** you can turn off one of the two chiplets:

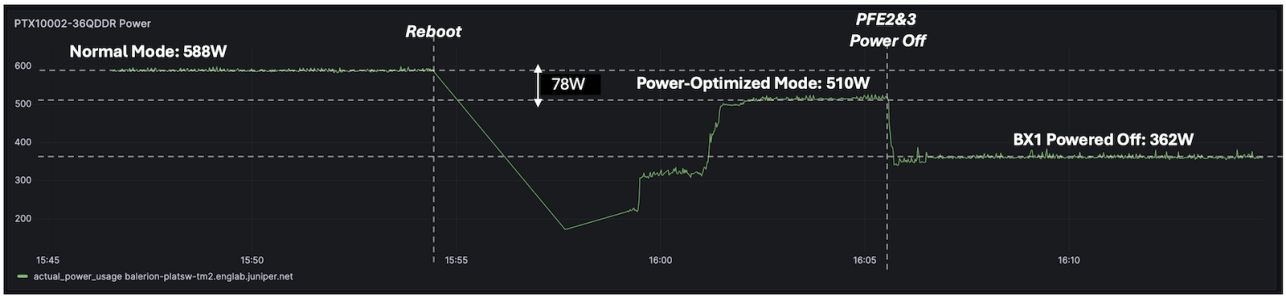


Figure 18: Enabling both methods

1.4.5 Best Practice

In the diagram below, we describe a case where 18 ports are initially connected "randomly" and where a smart repositioning of the interfaces on BX0 allows powering off the second BX1.

It's particularly relevant when you consider a PAYG approach.

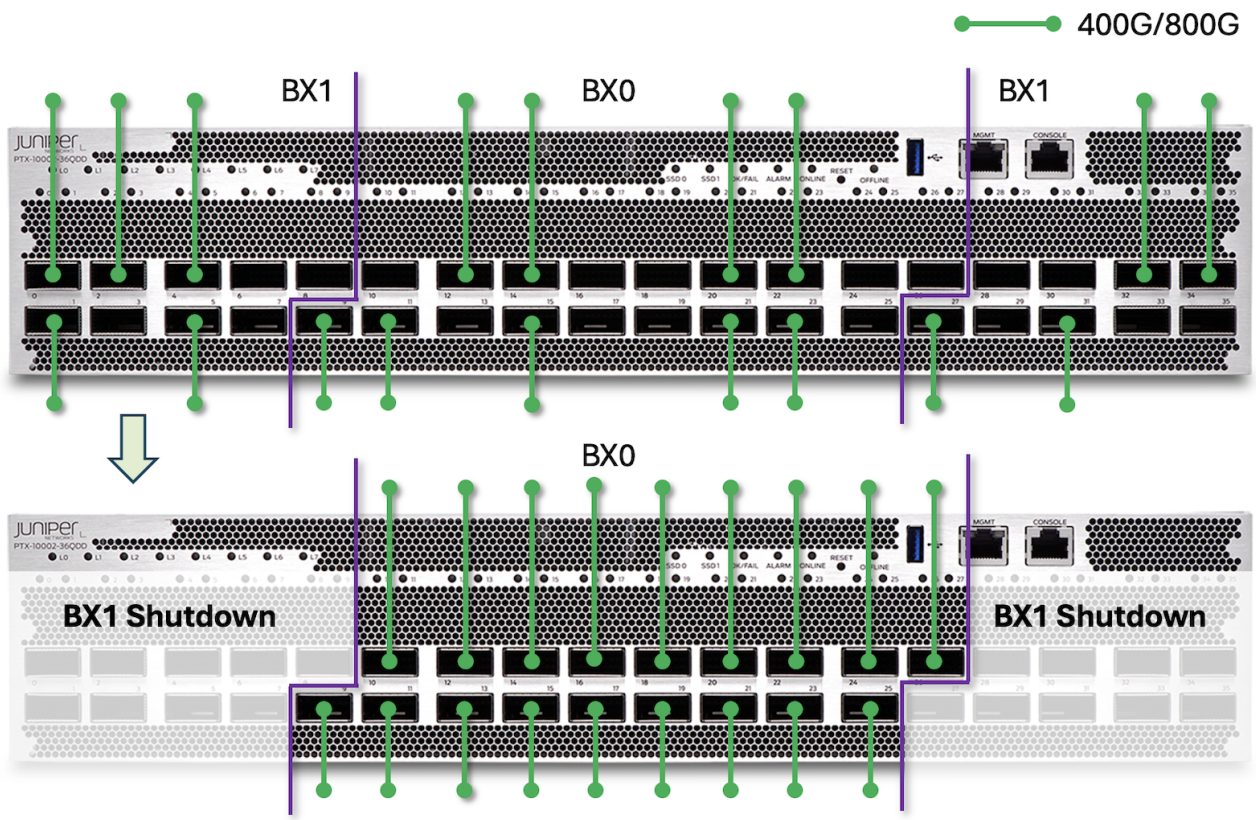


Figure 19: Optimal Port Position to Allow BX1 Power Off

The logic here is very simple: it's recommended to populate the pluggable optics in ports et-0/0/9 to et-0/0/26 first, that way we can power off half of the PFEs.

1.5 Conclusion

The PTX10002-36QDD is a highly compact and powerful router, offering unparalleled 800GigE and 400GigE port density in a 2RU form-factor. Built on a single Express 5 silicon chip with an optimized design, it delivers exceptional energy efficiency while also providing a comprehensive and continually expanding set of features and protocols.

This makes it an ideal choice for a wide range of applications, including core, peering, aggregation, data center edge, data center interconnect, and AI/ML clusters.

1.6 Useful links

- [The Feed] Introducing the PTX10002-36QDD Packet Transport Router: https://www.juniper.net/us/en/the-feed/topics/400g-and-800g/introducing-the-ptx10002-36qdd-packet-transport-router.html
- PTX10002-36QDD Datasheet: https://www.juniper.net/us/en/products/routers/ptx-series/ptx10002-fixed-configuration-packet-transport-routers-datasheet.html
- PTX10002-36QDD Specifications: https://www.juniper.net/us/en/products/routers/ptx-series/ptx10002-36qdd-packet-transport-router/specs.html
- Introduction to Express 5 TechPost blog: https://techpost.hpe.com/articles/express-5-overview/article
- FIB Compression blog: https://techpost.hpe.com/articles/ptx-fib-compression/article

1.7 Glossary

- ASIC: Application-Specific Integrated Circuit
- BIER: Multicast using Bit Index Explicit Replication
- BXX: Express 5 Package made of two X-chiplets back-to-back
- CLI: Command Line Interface
- CPU: Central Processor Unit
- DCI: DataCenter Interconnect
- DRAM: Dynamic Random Access Memory
- FIB: Forwarding Information Base
- FPGA: Field Programmable Gate Arrays
- FPC: Flexible PIC Concentrator
- GigE: Gigabit Ethernet
- HBM: High-Bandwidth Memory
- HQoS: Hierarchical Quality of Service
- INT-MD: Inband Network Telemetry Metadata
- LC: Little/Light Connector
- MACsec: Media Access Control Security
- MPLS: Multi-Protocol Label Switching
- MPO: Multi-Fibre Push-On/Off (Connector)
- PIC: Port Interface Card
- PPS: Packet Per Second
- PSM/PSU: Power Supply Module/Unit
- PTP: Precision Time Protocol
- QDD: QSFP Double Density
- RE: Routing Engine
- RGB: Reverse GearBox
- RIB: Routing Information Base
- SerDes : Serializer/Deserializer
- SoC: System on the Chip
- SRv6: Segment Routing IPv6
- SSD: Solid State Drive
- Sync-E: Synchronous Ethernet
- TPM: Trust Platform Module
- USB: Universal Serial Bus
- XSR: Extra Short Reach (SerDes)

1.8 Acknowledgments

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